Transport Management System - Detailed User Guide

Login Information

By default, the system does not require a login. If access control is introduced, usernames and passwords must be documented here.

System Purpose

This system is a command-line Java application designed to help logistics managers efficiently create, monitor, and manage transports between suppliers and branches using drivers and trucks. It includes full CRUD (Create, Read, Update, Delete) capabilities for all major components such as branches, drivers, trucks, and transport routes.

System Launch Instructions

1. Open a terminal window.
2. Navigate to the project directory.
3. Run the system using the following command:

`java -jar release/adss2025_v01.jar`

4. The application starts with a prompt: 'Load mock data? (y/n)'.

- Type 'y' to initialize mock data (sample branches, trucks, drivers, and suppliers).
- Type 'n' to start from scratch.

Main Menu Walkthrough

The main menu includes:

- Manage Transport Areas: Connect suppliers and branches into transport zones.
- Manage Transports: Create, view, and delete logistics operations.
- Manage Trucks: Maintain vehicle records.
- Manage Drivers: Add or update drivers and their license types.
- Manage Branches: Track branch locations and contact info.
- Exit: Close the program safely.
- Load Temporary Mock Data: Reinitialize demo data at any point.

Creating a Transport

To create a transport:

1. Choose 'Manage Transports' from the main menu.
2. Select 'Create transport'.
3. Enter a planned date and time for delivery.
4. Choose a source branch.
5. Select items from inventory to include in the transport.
6. System will attempt to find routes and available trucks/drivers.
7. You will be prompted to pick from valid trucks and drivers.
8. Upon success, a full transport is created and listed.

Drivers and Trucks

Each driver must be assigned a license type (B, C, C1, etc.).

Each truck requires a specific license type to operate.

The system only allows pairing a driver with a truck they are qualified to drive.

Mock Data Description

The mock data includes:

Branches

Three branches are added:

- **Branch1** – Address: "Address1", Phone: "+972 5412311"
- **Branch2** – Address: "Address2", Phone: "+972 5488899"
- **Branch3** – Address: "Address3", Phone: "+972 5412988"

Drivers

Three drivers with different license types:

- **John Doe** – ID: "123456789", Phone: "+972 5412311", License: "C"
- **Sam Wilson** – ID: "987654321", Phone: "+972 5488899", License: "B"
- **Jane Smith** – ID: "456789123", Phone: "+972 5412988", License: "C1"

Trucks

Three trucks with different capacities and required license types:

- **Truck 5498875** – Weight: 3500kg, Max load: 5000kg, License: "C1", Manufacturer: "Toyota"
- **Truck 5491176** – Weight: 4000kg, Max load: 6000kg, License: "C", Manufacturer: "Ford"
- **Truck 5491177** – Weight: 4500kg, Max load: 7000kg, License: "C1", Manufacturer: "Mercedes"

Suppliers

Three suppliers located at different addresses:

- **Supplier1** – Address: "Address1", Phone: "+972 77788855"
- **Supplier2** – Address: "Address2", Phone: "+972 77788856"
- **Supplier3** – Address: "Address3", Phone: "+972 77788857"

Transport Areas

Two transport areas are created, each associating existing branches and suppliers:

- **Area1** – Includes all created branches and suppliers
- **Area2** – Includes a subset (first two) of branches and suppliers

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Transport Planning

- A set of stock items is retrieved based on a mock search (with placeholder query "").
- A **transport route** is planned for the next day, using the selected stock items.
- The system finds an available truck and a driver with a matching license for the planned date.
- Finally, a **transport** is created linking the planned time, selected truck and driver, and the created transport routes.

Error Handling and Tips

The program uses Helper methods to validate user input. If an entry is invalid, the user is prompted again.

- Do not leave inputs blank.
- Follow format for dates and numbers.
- Ensure there are available drivers and trucks for the given transport date.

Extending the System

You can extend functionality by:

- Adding transport history and tracking.
- Introducing login/authentication features.
- Adding GUI components for ease of use.